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isk-Annotated Spatial Scene Graph for Dynamic Risk Assessment in Autonomous Systems

Master Thesis TBD

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Abstract

Conventional risk assessment mechanisms are limited when the control strategy or operating environment of an autonomous system changes after deployment. This limitation is particularly relevant for systems operating in dynamic environments or modified through adaptive software components and over-the-air updates.

This thesis contributes to the development of a dynamic risk assessment mechanism by extending an existing scene graph generation framework. The original FastAPI-based implementation is ported to ROS 2 to improve portability and integration with robotic systems. To support operation in larger environments, the scene graph generation process is extended through the integration of SLAM-based spatial data and loop closure detection. In addition, a probabilistic object representation is developed to account for uncertainty in object perception and to provide more robust information for risk assessment.

The thesis further refines the output of the vision-language model to produce bounded and consistent scene graph information. The proposed extensions are implemented with consideration for near-real-time operation, enabling the generated risk-annotated scene graph to provide structured environmental context during system operation.

The framework is evaluated using multiple datasets, including a manually generated dataset collected with a Unitree Go1 robot in a university environment. Overall, this work improves the scalability, portability, and robustness of spatial risk-annotated scene graph generation and supports the development of dynamic risk assessment mechanisms for autonomous systems.

Keywords: dynamic risk assessment, risk-annotated scene graph, ROS 2, SLAM, loop closure detection, probabilistic object representation, vision-language model

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1. Introduction

1.1. Motivation

Robotic systems are increasingly deployed across industrial, service, medical, and consumer domains. According to the International Federation of Robotics, annual installations of industrial robots increased from approximately 221,000 units in 2014 to 542,000 units in 2024, indicating that global robot demand in factories has roughly doubled over the last decade[1]. In the same period, the global operational stock of industrial robots increased from approximately 1.47 million units to 4.66 million units[1]. This development shows that robots are becoming a more widespread component of modern production and service environments.

At the same time, robotic systems are increasingly influenced by advances in artificial intelligence. The Stanford AI Index reports that 78% of organizations used AI in 2024, compared with 55% in the previous year [2].In addition, global private investment in generative AI reached USD 33.9 billion in 2024. The International Federation of Robotics also identifies physical, analytic, and generative AI as key technological trends in robotics. These developments indicate that future robotic systems will increasingly combine physical operation with adaptive, data-driven, and software-defined capabilities[1].

This combination introduces new challenges for safety and risk assessment. Conventional risk assessment methods are typically performed before deployment and rely on assumptions about the system design, intended use, operating environment, and known hazards [3]. Although such methods remain essential, they are limited when autonomous systems operate in dynamic environments, receive over-the-air software updates, or use adaptive AI-based components. This creates a need for dynamic risk assessment mechanisms that can incorporate the current operational context through structured representations of relevant objects, spatial relationships, and risk-related information while operating at near real time speed. A spatial risk-annotated scene graph is therefore motivated as a means of providing environmental context and situational awareness to support risk assessment during system operation.

1.2. Relevance

Dynamic risk assessment requires more than the detection of individual objects. In autonomous systems, risk depends not only on whether an object is present, but also on its position, spatial relationships, uncertainty, and relevance to the current operational context. A scene graph provides a suitable representation for this purpose because it organizes perceived objects, spatial relations, and semantic information in a structured and machine-readable form.

With the increasing number and variety of autonomous systems, portability and scalability become important requirements for risk assessment mechanisms. Such mechanisms should not be limited to a specific robot, sensor setup, or software interface, but should be reusable across different autonomous platforms. At the same time, the scene graph must be continuously updated as the environment changes and should be scalable to larger spatial operating areas by incorporating information from multiple observations over time.

Uncertainty handling and runtime performance are also essential for practical deployment. Since object detection, localization, and classification can contain errors, a probabilistic object representation can provide a more robust basis for risk analysis. Furthermore, dynamic risk assessment is only useful if environmental context is available during operation, which makes real-time or near-real-time scene graph generation a central requirement. Therefore, the relevance of this work lies in improving the practical applicability of risk-annotated scene graph generation for autonomous systems by addressing portability, scalability, uncertainty representation, and near-real-time operation.

1.3. Goal of the Thesis

The goal of this thesis is to extend an existing risk-annotated scene graph generation framework so that it can better support dynamic risk assessment in robotic systems. The work focuses on improving the framework in terms of portability, scalability, uncertainty handling, and near-real-time applicability.

First, the existing FastAPI-based implementation is ported to ROS 2 to improve integration with robotic software architectures. This allows the framework to communicate more directly with perception, localization, mapping, and planning components used in autonomous robotic systems. Second, the scene graph generation process is extended using SLAM-based spatial data and loop closure detection in order to support larger environments and maintain spatially consistent information over time.

Furthermore, a probabilistic object representation is developed to aid the risk analysis mechanism. The output of the vision-language model is also refined to produce bounded and consistent scene graph information. Since dynamic risk assessment requires timely environmental context, near-real-time performance is considered throughout the design and implementation of the extended framework.

Finally, the resulting framework is evaluated using multiple datasets, including a dataset collected with a Unitree Go1 robot in a university environment.

2. Theoretical Background

2.1. Autonomous Systems and Safety

Autonomous systems (AS) can be understood as systems that are capable of making decisions independently from direct human control [4]. In robotic systems, this autonomy is typically achieved through a combination of sensing, perception, planning, decision-making, and actuation. For example, an autonomous mobile robot may perceive obstacles, select an alternative path, and continue its task without requiring continuous human input. This ability allows autonomous robots to operate more flexibly in dynamic and partially unknown environments. However, it also makes safety assurance more complex, because the system behaviour is no longer determined only by predefined commands, but also by runtime perception, environmental conditions, and decision-making processes.

Safety is a fundamental requirement in AS because robots interact with humans, physical objects, infrastructure, and the surrounding environment. In dependability theory, safety is commonly defined as the absence of catastrophic consequences for users and the environment [5]. In the context of autonomous robotic systems, safety therefore refers not only to whether the robot performs its intended task, but also to whether it performs the task without creating unacceptable risk. This distinction is important because a robot may be functionally successful while still being unsafe. For example, a robot arm in a shared workspace may complete a pick-and-place task correctly, but the system cannot be considered safe if its motion exposes a nearby human operator to an unacceptable collision risk.

From a risk assessment perspective, safety is closely related to the identification of hazards, the estimation of risk, and the implementation of suitable risk reduction measures [3]. In robotic applications, this requires considering both the severity of possible harm and the likelihood that a hazardous situation may occur. Robot-specific safety standards further define requirements and protective measures for industrial and collaborative robotic systems [6]–[8]. However, safety in AS is strongly dependent on operational context. A motion, speed, or decision that is safe in one situation may become hazardous when a human enters the workspace, when an object is misplaced, or when perception uncertainty increases. Therefore, safety in autonomous robotic systems must be understood as a property that depends on both the technical system and the changing environment in which it operates.

2.1.1. Human-Robot Collaboration

Industry 5.0 is shifting industrial development toward a more human-centric, sustainable, and resilient model, while continuing to support productivity and worker well-being [9]. Within this perspective, Human-Robot Collaboration (HRC) becomes an important requirement for using robotic technologies to support and empower workers rather than simply replacing human labour [9]. Collaborative robotic systems can assist humans in repetitive, strenuous, or hazardous tasks, thereby improving workplace flexibility, ergonomics, and the distribution of work between humans and machines. In industrial settings, HRC has also been discussed as a response to skill gaps, skilled-worker retention, and the need to make manufacturing work more attractive to younger generations [10]. This makes HRC a relevant foundation for future autonomous systems, where robots increasingly need to interpret human presence, environmental changes, and task-specific context during operation.

However, HRC also introduces important challenges in industrial settings. Industrial HRC must address three closely related aspects: ensuring human safety, maintaining trust in automation, and preserving productivity [10]. Among these, human safety remains a primary concern because close physical interaction increases the possibility of unintended human-robot collisions, unexpected robot motion, or interruptions in the robot's behaviour. Such situations can increase the risk of injury and may negatively affect the human operator's trust in the system. At the same time, safety measures must be balanced with productivity requirements, since overly restrictive measures such as excessive speed reductions, large separation distances, or frequent safety stops can reduce system efficiency and interrupt the intended collaborative workflow [10]. The relationship between safety, trust in automation, and productivity is illustrated in Figure 2.1. This trade-off shows that HRC cannot be addressed only as a productivity or automation problem; it also requires systematic risk assessment methods that can identify hazardous situations and define suitable risk reduction measures.

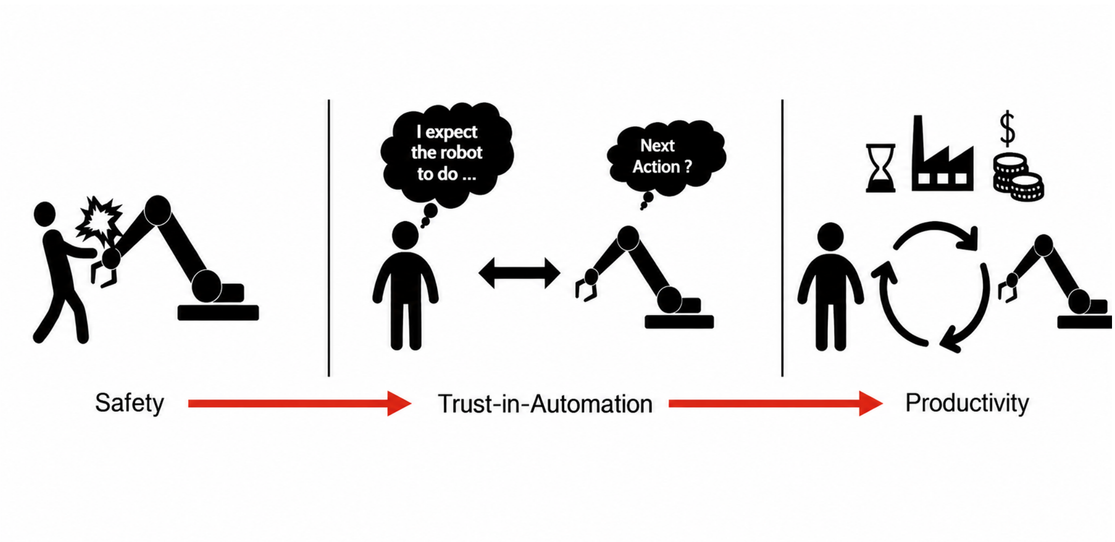

Figure 2.1.: Main challenges in HRC[10].

2.1.2. Conventional Risk Assessment and Risk Reduction

Conventional risk assessment in robotic systems is generally performed before deployment and is based on the intended use, foreseeable misuse, system limits, operating environment, and known hazards of the system. According to ISO 12100, the risk assessment process includes hazard identification, risk estimation, and risk evaluation, followed by the selection of suitable risk reduction measures [3]. In the context of industrial and collaborative robots, this process is supported by robot-specific safety standards that define requirements for robot systems, collaborative applications, and protective measures [6]–[8].

Risk reduction is usually implemented through a combination of inherently safe design, technical protective measures, and information for use. In HRC, this can include limiting robot speed and force, defining safe separation distances, using protective stops, and applying physical or electronic safeguards around the robot workspace. These measures are essential because they provide a systematic way to reduce the likelihood or severity of hazardous human-robot interactions before the system is put into operation.

However, conventional risk assessment is mainly based on assumptions made during design and deployment. These assumptions include expected human behaviour, known object configurations, defined workspaces, and predefined task conditions. As a result, conventional risk assessment is effective for identifying known and reasonably foreseeable hazards, but it becomes limited when the operating context changes significantly during runtime. This limitation is especially relevant for autonomous robotic systems, where the robot may operate in dynamic environments and make decisions based on sensor data and perception results.

2.1.3. Existing Safety Measures in HRC

In industrial HRC, risk reduction strategies are commonly implemented through two main approaches: the use of collaborative robots and the use of physical or electronic safeguarding measures [10]. Collaborative robots, also referred to as cobots, are designed to operate in direct cooperation with humans within a defined shared workspace. Their safety is supported through hardware and software features such as reduced momentum, force sensing, back-drivable actuators, and protective stop functions that interrupt robot motion when unexpected contact or resistance is detected.

The safeguarding-based approach focuses on monitoring the robot workspace so that robot motion can be adapted or stopped before hazardous human-robot collisions occur. In practice, this is achieved using physical and electronic safeguarding devices that define protected zones around the robot work cell. Examples include safeguarded perimeter door switches, light curtains, pressure-sensitive floors, 2-D scanning LiDAR, and 3-D safety vision systems. These mechanisms are important for reducing the risk of human injury, but they often rely on predefined safety zones, fixed distance thresholds, or conservative stopping rules [10].

A common method used for workspace monitoring is Speed and Separation Monitoring (SSM). In SSM, the distance between the human and the robot is monitored continuously, and the robot speed is reduced or stopped when the separation distance becomes too small. In a static SSM approach, the monitored zones are usually predefined before operation. For example, a robot may operate at full speed when no human is detected in the outer zone, slow down when a human enters a warning zone, and stop when the human enters a protective stop zone[10]. This approach is simple and practical, but it can be conservative because the safety zones are not always adapted to the actual robot motion, human motion, task state, or sensor uncertainty.

Figure 2.2.: Illustrative representation of safety zones in a collaborative robot workspace.

Dynamic SSM addresses this limitation by adapting the protective separation distance or the robot speed according to the current operating situation. Instead of relying only on fixed zones, dynamic SSM considers factors such as the actual distance between the human and the robot, robot velocity, human motion, stopping time, and uncertainty in the sensed position of obstacles or operators. Marvel discusses SSM performance in terms of both safety and productivity, representing collision potential through separation metrics and sensor uncertainty [11]. This shows that SSM is not only a binary stop-or-run mechanism, but also a method whose effectiveness depends on how accurately the system evaluates separation, uncertainty, and the productivity cost of safety actions.

Although dynamic SSM provides a more flexible basis for HRC safety than purely static safety zones, it also requires reliable perception, continuous monitoring, and suitable runtime decision-making. Therefore, existing safety measures show the importance of moving from fixed protective assumptions toward context-aware risk assessment mechanisms that can consider the actual operational situation of the robot, humans, and surrounding environment.

2.1.4. Impact of Artificial Intelligence

The previous subsections show that conventional risk assessment and many existing HRC safety measures are based on pre-deployment assumptions, such as intended use, expected human behaviour, known object configurations, predefined workspaces, fixed safety zones, and foreseeable hazardous scenarios. These assumptions are necessary for systematic risk assessment, but they become less reliable when autonomous robots operate in changing environments. Human positions, object locations, obstacles, task conditions, and spatial relationships may vary during operation, meaning that a risk analysis performed before deployment may no longer fully represent the actual operating situation.

The integration of artificial intelligence into robotic perception, planning, or control further increases this limitation. AI-based components can enable object detection, scene interpretation, motion prediction, task planning, and adaptive decision-making. While these capabilities improve flexibility and autonomy, they also make robot behaviour harder to predict completely during design-time safety analysis. Safety assurance of AI-based systems is complicated by dataset dependence, black-box behaviour, explainability, and lifecycle assurance [12]. In robotic systems, this means that perception errors, low-confidence predictions, distribution shifts, previously unseen objects, changing human behaviour, or software updates after deployment can affect system decisions.

As a result, conventional risk assessment should be understood as a necessary baseline rather than a complete solution for adaptive autonomous robotic systems. If the deployed system operates under conditions that were not fully represented during the original assessment, the selected safety measures may become insufficient. Therefore, dynamic and context-aware risk assessment is required to incorporate runtime information such as robot state, human presence, object locations, spatial relationships, perception uncertainty, and environmental changes. Automated and continuous risk assessment has been proposed for software-defined robotic systems, where changes in software or system behaviour may require risk assessment to be updated during operation [13].

At the same time, AI can also support dynamic risk assessment when its outputs are structured, bounded, and interpretable. AI-based perception and reasoning methods can help identify relevant objects, detect environmental changes, classify scene elements, and localise risk-relevant information around the robot. In this thesis, this motivates the use of a spatial risk-annotated scene graph, which can represent objects, spatial relations, uncertainty, and risk-related context in a machine-readable form to support dynamic risk assessment in autonomous robotic systems.

2.2. Section2

2.3. Section3

2.4. Section4

2.5. Section5

2.6. Section6

2.7. Section7

3. Methodology

4. Experiments

5. Conclusions

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A. Example appendix chapter

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